

ISHITA

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EDUCATION

B.Tech in Computer Science and Engineering

VIT Bhopal University

June 2027

CGPA: 8.75

Relevant courses: Data Structures and Algorithms, Machine Learning, Deep Learning, Database Management Systems, Object-Oriented Programming, Computer Networks, Operating Systems, Statistics for Data Science

Class 12th: 95.6% ; Class 10th: 97.6%

2020 - 2022

Modern Delhi Public School

TECHNICAL SKILLS

Programming languages: Python, Java

Machine learning: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, OpenCV

Data science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering

Tools & technologies: Git, Flask, MLflow, AWS, REST, SQL

PROJECTS

Deep News Summarizer – End-to-End NLP Pipeline for News Analysis

[GitHub](#)

- Designed and implemented an end-to-end NLP pipeline for news ingestion, preprocessing, and abstractive summarization using **Transformer-based models (BART, DistilBART, PEGASUS)**.
- Developed a multi-stage analysis workflow combining **KeyBERT** keyword extraction and **RoBERTa** sentiment classification to generate structured insights from news articles.
- Built a **Dockerized Streamlit application** with GPU-accelerated inference (**2s/article**) and support for processing long-form articles through automatic text chunking.

IntelliVoice – End-to-End Automatic Speech Recognition

[GitHub](#)

- Developed an end-to-end **speech-to-text system** using a **DeepSpeech2-inspired architecture** with Residual CNN and BiGRU layers, trained on a **2,000-sample LibriSpeech subset**.
- Built an audio processing pipeline using **PyTorch**, **TorchAudio**, and **MFCC feature extraction**, and trained the model with **CTC Loss** and **AdamW** optimization.
- Integrated **MLflow** for experiment tracking and model management, and evaluated transcription quality using **Word Error Rate (WER)** and **Character Error Rate (CER)**.

DWSIM Surrogate Model for Distillation Column

[GitHub](#)

- Generated **500+ simulation samples** using Latin Hypercube Sampling across 7 process variables by automating DWSIM distillation simulations with Python.
- Trained and benchmarked **4 ML models** to predict key outputs and achieved **95%+ accuracy** using Random Forest with highest R^2 and lowest MAE/RMSE.
- Reduced simulation time by **90–95%**, replacing iterative DWSIM runs with near-instant ML predictions.

CERTIFICATIONS

Applied Machine Learning in Python — University of Michigan

Dec 2024

Workshop on Containers and AIML on AWS Cloud — Cognizance, IIT Roorkee

Apr 2024

AWS Cloud Clubs Machine Learning Camper — United Latino Students Association

Mar 2024

ADDITIONAL INFORMATION

- Languages:** Fluent in English and Hindi; Basic proficiency in French
- Achievement:** Top 50 among 267 teams nationwide in Healthcare Hackathon organized by VIT Bhopal and Johns Hopkins University.
- Soft Skills:** Problem-solving, teamwork, leadership, communication, time management, adaptability